Explainable and Threshold-Aware Machine Learning for Customer Churn Prediction Using XGBoost

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***Abstract*—Customer churn prediction allows telecommunications providers to direct limited retention resources toward the subscribers who are most likely to cancel. This paper presents an end-to-end machine-learning workflow for binary churn classification on the IBM Telco Customer Churn dataset, combining model comparison, threshold-aware evaluation, probability calibration, and SHapley Additive exPlanations (SHAP). Logistic Regression, AdaBoost, XGBoost, and Gaussian Naive Bayes are compared, and XGBoost, which attains the highest ROC-AUC (0.8378), is selected for detailed analysis. At a decision threshold of 0.30, the model achieves a recall of 0.7647, a precision of 0.5044, an F1-score of 0.6079, and an accuracy of 0.7377 on a held-out test set of 1,407 customers, detecting 286 of the 374 churners. Global and local SHAP analyses identify contract type, internet service, tenure, online security, and billing-related features as the main contributors to the model output. The trained model is deployed in a Streamlit application that returns a churn probability and a risk label for an individual customer. The results illustrate how the decision threshold trades recall against precision, and the paper emphasizes that SHAP attributions describe model behavior rather than causal effects.**

***Index Terms*—Customer churn, telecommunications, XGBoost, machine learning, SHAP, explainable artificial intelligence, decision threshold, Streamlit.**

I. INTRODUCTION

Customer churn occurs when a subscriber discontinues a service. In subscription-based telecommunications, churn prediction is useful because retention teams have limited time and budgets and cannot offer the same intervention to every customer. A predictive model can rank customers by estimated churn risk, while an operating threshold converts scores into an action list. The resulting system must therefore be assessed not only by a single accuracy value but also by its ability to identify churners, the number of false alarms it generates, and the reliability of its probabilities.

Tree ensembles can capture nonlinear relationships and interactions among customer attributes. XGBoost is a scalable gradient-boosting system that combines regularized tree learning with an efficient optimization procedure [1]. However, ensemble predictions are not directly transparent to most business users. SHAP provides a unified framework for assigning feature contributions to individual predictions and summarizing those contributions across a dataset [2].

This paper develops an end-to-end churn-prediction workflow on the IBM Telco Customer Churn dataset [3]. The workflow jointly considers model discrimination, decision-threshold trade-offs, probability calibration, global feature importance, and local explanations, and it is packaged as a lightweight prototype that lets an analyst inspect a customer's predicted risk and the features that drive it. The main contributions are as follows.

- A comparison of four classifiers (Logistic Regression, AdaBoost, XGBoost, and Gaussian Naive Bayes) under a common preprocessing pipeline and a common set of evaluation metrics.
- A threshold-aware evaluation of XGBoost that reports precision, recall, F1-score, and confusion-matrix counts at a recall-oriented operating point instead of relying on the default 0.50 cutoff.
- A global and local SHAP analysis that relates the model output to contract, service, tenure, and billing attributes.
- A Streamlit prototype that scores individual customers using the trained model.

The contribution is an integrated, practice-oriented workflow rather than a new algorithm; the learning methods and SHAP are used as published. The remainder of the paper is organized as follows. Section II reviews related work, Section III describes the data, Section IV presents the methodology, Section V reports the results, Section VI discusses them, Section VII lists limitations, and Section VIII concludes.

II. RELATED WORK

Churn prediction has been studied using statistical methods, conventional machine learning, ensembles, and more recent deep-learning approaches. Reviews of telecom churn research describe variation in datasets, model choices, metrics, and validation procedures, which makes direct comparison across papers difficult [4], [5]. Results obtained on different customer populations or under different split strategies should therefore not be treated as directly comparable.

Chen and Guestrin introduced XGBoost as a scalable tree-boosting system and described the regularization and engineering optimizations that support efficient learning [1]. In this study, XGBoost is evaluated alongside Logistic Regression, AdaBoost, and Gaussian Naive Bayes, which provide linear, boosting-based, and probabilistic baselines. Their relative performance is measured on identical data partitions rather than inferred from algorithm type.

Lundberg and Lee proposed SHAP, a framework grounded in Shapley-value attribution, for explaining individual model outputs [2]. In this paper, SHAP is used for global summaries and local explanations. SHAP values are interpreted as contributions to the model output under the explainer's output scale and background distribution; they are not causal effects. The review literature also highlights explainability as an important complement to predictive performance in churn analysis [4], [5].

III. DATASET AND PREPROCESSING

A. Dataset

The study uses the IBM Telco Customer Churn sample dataset, which describes a fictional telecommunications provider and defines the churn field as whether a customer departed within the last month [3]. The publicly distributed file contains 7,043 customer records and 21 columns covering customer demographics, service subscriptions, contract information, and billing attributes. The target variable is Churn, encoded as a binary label.

The predictors comprise demographic attributes (gender, SeniorCitizen, Partner, and Dependents); the derived categorical feature tenure_group; telephone and internet service options (PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies); contract and billing attributes (Contract, PaperlessBilling, and PaymentMethod); and the numerical variables MonthlyCharges and TotalCharges.

B. Data Preparation

MonthlyCharges and TotalCharges are retained as numerical inputs, and the categorical features are one-hot encoded, giving 50 model input columns. Preprocessing is fitted on the training partition only and then applied unchanged to the validation and test partitions. This avoids information leakage and ensures that the columns seen at inference time match the training schema. The trained XGBoost model is an ensemble of 100 boosted trees. Feature names in the figures of Section V correspond to the 50 transformed columns.

IV. METHODOLOGY

A. Candidate Models

Four classifiers are compared. Logistic Regression provides a linear baseline. Gaussian Naive Bayes assumes conditional independence of the features given the class, with Gaussian class-conditional distributions. AdaBoost is an adaptive boosting ensemble, and XGBoost is a gradient-boosted decision-tree ensemble with regularized objectives.

B. Training and Evaluation Protocol

The data are divided into training, validation, and test partitions. Model fitting, hyperparameter tuning, and threshold selection use the training and validation partitions only, and the test partition is reserved for the final evaluation. Threshold curves computed on the test partition are descriptive and are not used to tune the operating point.

C. Evaluation Metrics

Accuracy measures the fraction of correct predictions. Precision measures the fraction of predicted churners who actually churn, recall measures the fraction of actual churners that are detected, and the F1-score is the harmonic mean of precision and recall. ROC-AUC measures ranking discrimination across all thresholds, while average precision summarizes the precision–recall curve, which is informative when classes are imbalanced [7]. The Brier score [6] is the mean squared error between predicted probabilities and binary outcomes, with lower values being better. Confusion-matrix counts are reported to make the practical error trade-off visible.

D. Decision Threshold

A probability threshold converts a model score into a binary action. Lowering the threshold flags more customers as at risk, which usually increases recall and decreases precision. On the validation partition, the F1-score is maximized at a threshold of 0.33 (F1 = 0.6281). The operating point adopted for the prototype and for the main test evaluation is 0.30, which lies slightly below this value and therefore favors churn detection.

E. SHAP Explainability

SHAP is used to produce (i) a global ranking of features by mean absolute SHAP value, (ii) a summary (beeswarm) plot showing the distribution and direction of feature contributions, and (iii) a local waterfall explanation for a selected customer. The waterfall plot explains a single model output relative to the explainer's baseline and is expressed on the log-odds (raw margin) scale rather than on the probability scale.

F. Prototype Deployment

A Streamlit application serves as the prototype interface. It loads the persisted XGBoost model and the matching fitted preprocessor, accepts the attributes of a customer, applies the transformations used during training, and returns a churn probability together with a threshold-based risk label. The application is a demonstration interface and is not claimed to be production-ready.

V. EXPERIMENTAL RESULTS

A. Model Comparison

Table I compares the four candidate models on the test set at the default decision threshold of 0.50. XGBoost attains the highest ROC-AUC (0.8378), followed closely by AdaBoost (0.8344) and Logistic Regression (0.8306), so the differences among these three models are small. Gaussian Naive Bayes achieves the highest recall (0.8209) and a marginally higher F1-score than XGBoost (0.5842 versus 0.5802), but it has the lowest precision (0.4535), accuracy (0.6894), and ROC-AUC (0.8078). Because ROC-AUC measures ranking quality independently of any single threshold, XGBoost is selected for further analysis.

TABLE I
PERFORMANCE OF CANDIDATE MODELS ON THE TEST SET (THRESHOLD 0.50)

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.7918	0.6401	0.4947	0.5581	0.8306
AdaBoost	0.7953	0.6396	0.5267	0.5777	0.8344
XGBoost	0.7953	0.6378	0.5321	0.5802	0.8378
Gaussian Naive Bayes	0.6894	0.4535	0.8209	0.5842	0.8078

B. XGBoost at Threshold 0.30

Table II and Fig. 1 summarize the performance of the XGBoost model at the 0.30 threshold. Threshold-independent measures are favorable: ROC-AUC is 0.8378 and average

precision is 0.6485, as shown in Fig. 3, and the Brier score is 0.1389.

TABLE II
XGBOOST TEST-SET PERFORMANCE AT THRESHOLD 0.30

Metric	Reported value
Accuracy	0.737740
Precision	0.504409
Recall	0.764706
F1-score	0.607864
ROC-AUC	0.837766
Average precision	0.648510
Brier score	0.138919
True negatives / false positives	752 / 281
False negatives / true positives	88 / 286

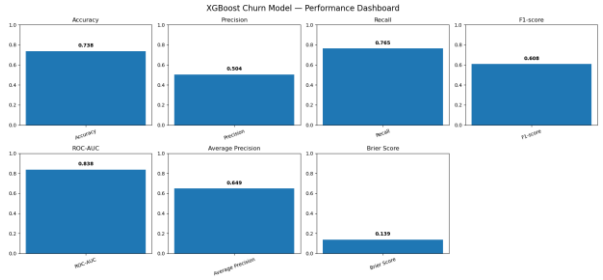


Fig. 1. XGBoost evaluation dashboard at the 0.30 decision threshold.

The confusion matrix in Fig. 2 contains 752 true negatives, 281 false positives, 88 false negatives, and 286 true positives over the 1,407 test customers. The model therefore detects 286 of the 374 actual churners (76.5%) and misses 88, while 281 non-churners are flagged as at risk. In total, 567 customers (40.3% of the test set) are flagged; 50.4% of them are genuine churners, compared with a base churn rate of 26.6% in the test set, a lift of approximately 1.9.

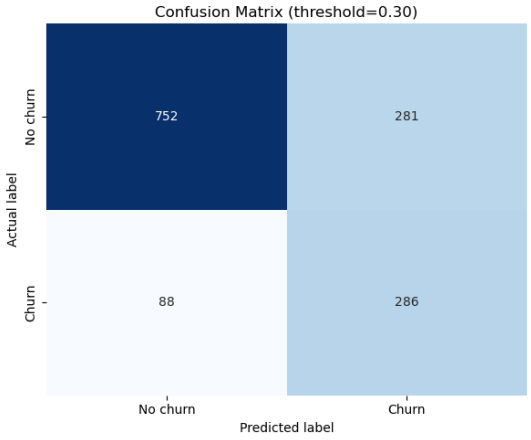


Fig. 2. Confusion matrix at the 0.30 decision threshold.

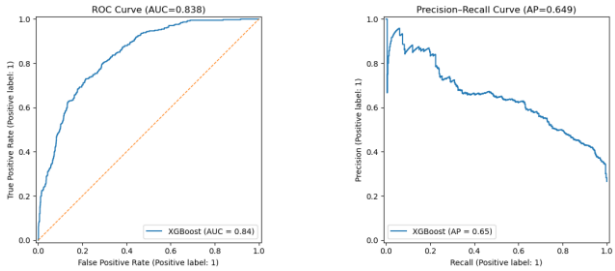


Fig. 3. ROC and precision–recall curves; ROC-AUC = 0.838 and average precision ≈ 0.649.

Fig. 4 shows how the threshold affects precision, recall, F1-score, and accuracy. Relative to the default 0.50 threshold in Table I, the 0.30 operating point raises recall from 0.5321 to 0.7647 and F1-score from 0.5802 to 0.6079, while precision falls from 0.6378 to 0.5044 and accuracy falls from 0.7953 to 0.7377. This operating point favors churn detection at the cost of a substantial number of false positives.

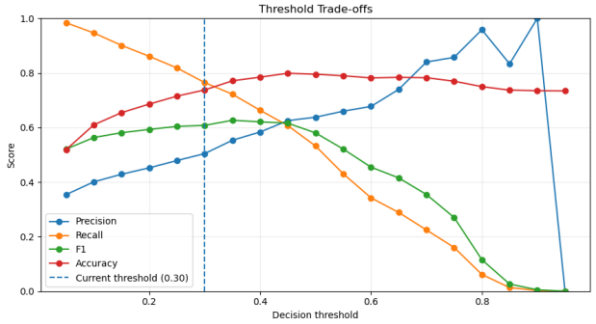


Fig. 4. Threshold trade-offs for precision, recall, F1-score, and accuracy. The dashed line marks the 0.30 operating point.

The calibration plot in Fig. 5 shows that the predicted probabilities follow the observed churn frequencies reasonably closely, with greater variability at high predicted probabilities, where fewer observations are available. The Brier score is approximately 0.139.

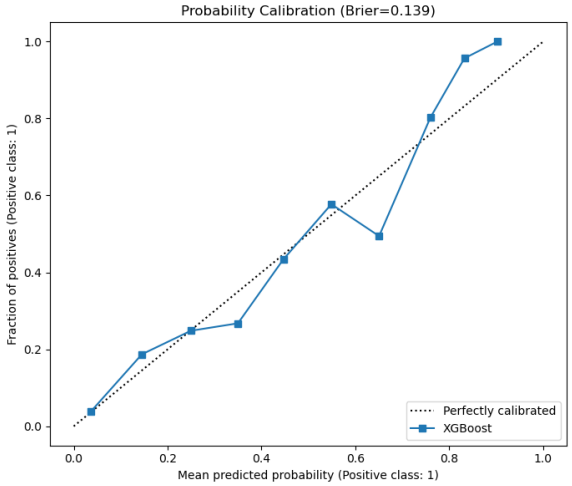


Fig. 5. Probability calibration plot; Brier score ≈ 0.139.

C. Feature Importance and Explanations

The built-in XGBoost importance chart in Fig. 6 ranks OnlineSecurity categories, contract type, tenure group, technical support, internet service, payment method, and charges among the most influential transformed inputs.

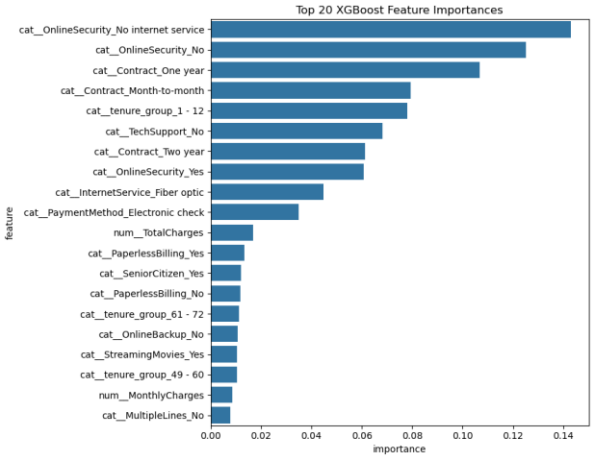


Fig. 6. Top 20 built-in XGBoost feature importances.

The SHAP global plot in Fig. 7 ranks month-to-month contract, fiber-optic internet service, two-year contract, TotalCharges, and the 1–12 month tenure group among the leading contributors by mean absolute SHAP value. The summary plot in Fig. 8 further shows that the direction of the contribution differs across feature values and categories. Because the contract categories are distinct binary indicators produced by one-hot encoding, each must be interpreted with reference to its original category.

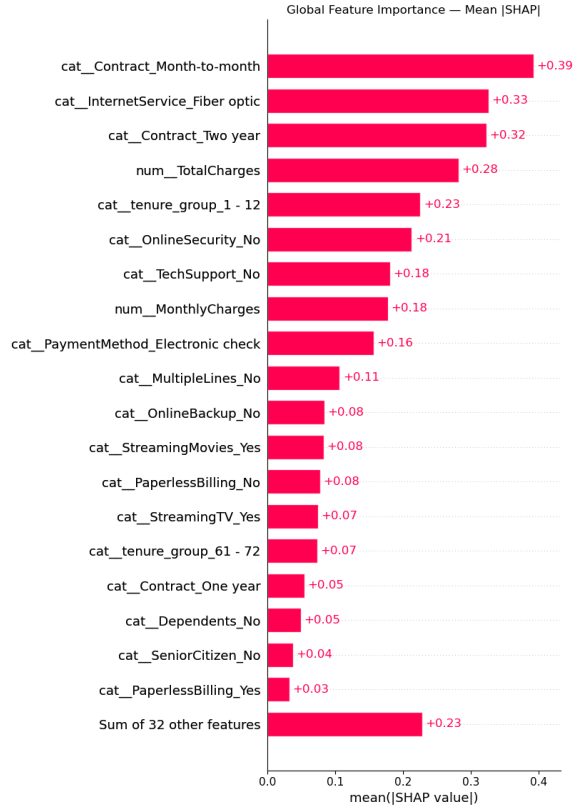


Fig. 7. Global feature importance using mean absolute SHAP values.

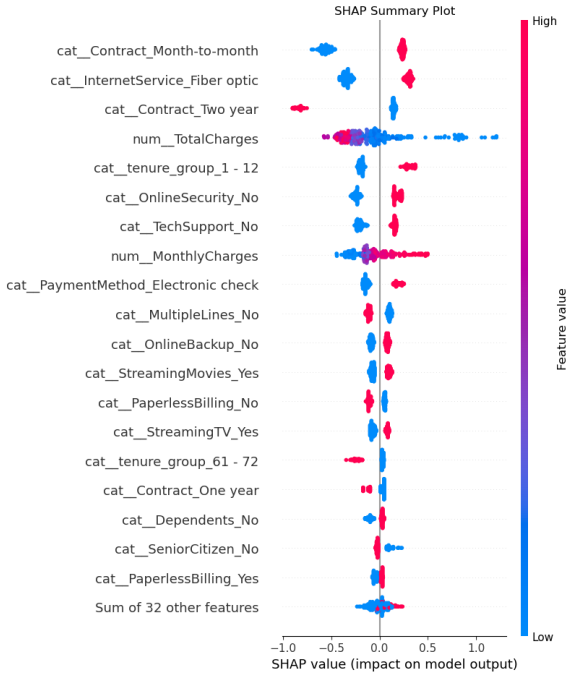


Fig. 8. SHAP summary plot for the XGBoost model.

The waterfall plot in Fig. 9 shows how feature contributions combine to move a selected observation away from the baseline model output. In this example, contract-related indicators and service-related attributes contribute substantially to the displayed output. A single-customer explanation is illustrative and is not a population-level conclusion. Fig. 10 shows the first tree of the 100-tree ensemble, truncated for display, to illustrate the structure of the model.

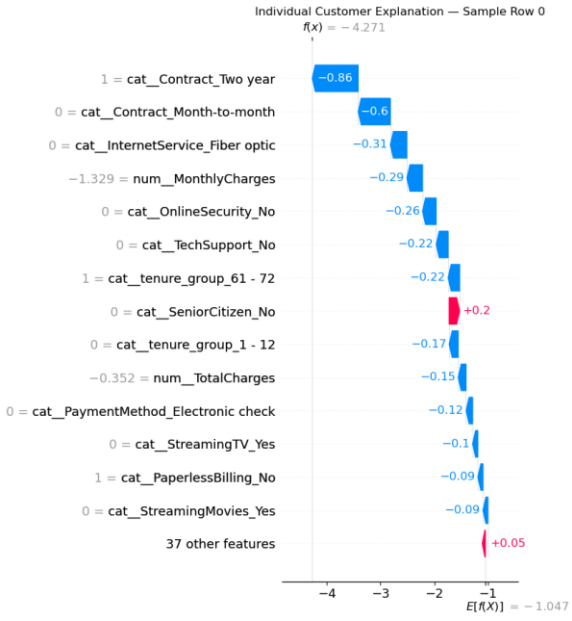


Fig. 9. Local SHAP waterfall explanation for one example customer.

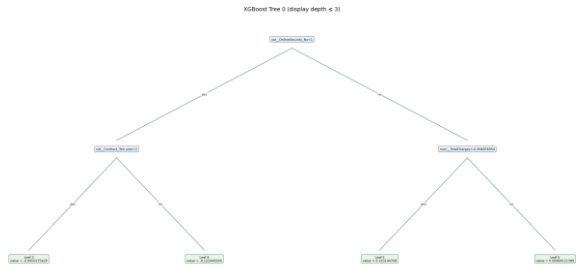


Fig. 10. First boosted tree, truncated for display; one of 100 trees in the ensemble.

VI. DISCUSSION

The results indicate that the tuned XGBoost model separates churners from non-churners well in ranking terms, with a ROC-AUC of 0.838. Lowering the threshold from 0.50 to 0.30 increases recall from 0.532 to 0.765 while accuracy decreases, because more customers are classified as at risk. This decrease is a consequence of choosing an operating point that prioritizes detection rather than a deficiency of the model. The appropriate threshold depends on campaign capacity, the cost of contacting customers who would not have left, the value of retained customers, and the effectiveness of the intervention.

The lift of approximately 1.9 at the 0.30 operating point shows that the flagged group is considerably richer in churners than the customer base as a whole, so a retention team that contacts only the flagged customers reaches most churners with fewer contacts than random targeting. The false-positive workload remains substantial, however, and a campaign with a low contact cost can tolerate it more readily than one with a high cost.

SHAP adds an interpretation layer to the predictive model. Contract type, service configuration, tenure group, and billing-related variables appear influential in the explanations. These patterns can guide further analysis, but they should not be treated as causal relationships or as universal churn rules. The dataset describes a fictional provider and may not represent the customer behavior of a particular operator, region, or market.

VII. LIMITATIONS

- **Data:** the dataset describes a fictional provider and may not generalize to live telecom operations.
- **Evaluation design:** results come from a single train/validation/test split without cross-validation or confidence intervals, so small differences between models, such as the F1-scores of XGBoost and Gaussian Naive Bayes, should not be regarded as established.
- **Threshold:** the operating point of 0.30 differs from the validation F1-optimal threshold of 0.33, and test performance at 0.33 is not reported. A cost-based threshold would require retention-cost estimates that are not available here.
- **Calibration:** calibration is assessed on one test partition and is not compared with a baseline or recalibrated.
- **Explainability:** SHAP values are model-specific attributions, not causal explanations, and a single local example does not generalize.
- **Deployment:** the Streamlit prototype does not by itself establish security, latency, monitoring, or operational suitability.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a workflow for telecom customer churn prediction that combines supervised learning, threshold analysis, SHAP explanations, and a Streamlit prototype. Among four candidate classifiers, XGBoost achieved the highest ROC-AUC (0.8378). At a decision threshold of 0.30, it reached a recall of 0.7647, a precision of 0.5044, and an F1-score of 0.6079 on the test set. SHAP analysis associated contract type, internet service, tenure, and billing-related features with the model output, and the results showed that the choice of threshold, rather than accuracy alone, determines the practical value of a churn model.

Future work should (1) evaluate all candidate models under a common stratified cross-validation protocol and report means and standard deviations across folds; (2) select the decision threshold using an explicit retention-cost model; (3) assess calibration and expected business utility on an untouched test set; (4) test robustness across customer subgroups and alternative datasets; and (5) evaluate the prototype with retention staff before considering operational deployment.

ACKNOWLEDGMENT

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